

Technical Note:

Errors in the Analytic Photon Phase-Space Integrals and Physical Constants of MERADGEN 1.0

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Abstract

Four integrals in the analytically integrated branch of the MERADGEN 1.0 real-photon cross section routine `fsir.f` are incorrect. The faults are a transcription error in `aj47`, a dimensional error in `aj29`, a dimensional error in `aj28`, and a sign error in `aj51`. An apparent sign discrepancy in the logarithmic term of `aj28` which I have not had full time to analyze is reported as an open item requiring **further investigation** but is detailed here. Each fault is identified, located by line number in the original FORTRAN code, corrected and verified analytically and numerically against the integral definitions given in the same code, and ultimately corrected in my C++ port. All faulty terms carry an explicit factor of the electron mass squared and vanish in the ultra-relativistic limit, which is why comparison with the published ultra-relativistic results cannot detect them; they've been corrected here as a matter of being correct rather than a matter of corrupted calculations. I also note the constants corrected in my C++ port to be used for MolPol.

1 Scope and context

MERADGEN 1.0 [1] generates radiative events in doubly polarized Møller scattering. Its routine `fsir.f` evaluates the real-photon emission cross section, fully differential (`ikey=0`), integrated over the photon variable z (`ikey=1`), or integrated over z and t_1 (`ikey=2`, and `ikey=-1` for the infrared-subtracted finite part). The ultrarelativistic limit of this calculation is published in Ref. [2]. The exact expressions beyond the ultrarelativistic limit are published only as this source code; Ref. [3] cites MERADGEN 1.0 as their reference implementation (although I used the paper for general reference to their formulations).

The issues were found while porting MERADGEN to C++ for use in Møller polarimetry for the MOLLER experiment at Jefferson Lab. All line numbers in this note refer to the original Fortran source of MERADGEN 1.0, files `fsir.f` (857 lines) and `meradgen10.f` (860 lines), not to the C++ port.

1.1 Structure of the calculation

`fsir` returns `coer*(sr1+...+sr10)`, line 847. Each `srN` is a linear combination of 51 basis integrals `aj1-aj51` with polynomial coefficients. In the `ikey=0` block (lines 43–96) each `ajNN` is defined as an integrand; the `ikey=1` block (lines 97–148) gives the z -integrated forms; the `ikey=2` block (lines

149–343) gives closed forms integrated over z and t_1 . The **ikey=2** closed forms are, by construction, the **ikey=0** integrands integrated over the photon phase space [1, Eqs. (13), (16)]. This definitional relation is the basis of every check below. In particular, for every basis integral the **ikey=2** closed form must equal the **ikey=1** expression integrated over t_1 between the limits computed by the code (`meradgen10.f` lines 113–115):

$$\text{ajNN}^{(\text{ikey}=2)}(s, t, v) = \int_{t_1^{\min}}^{t_1^{\max}} \text{ajNN}^{(\text{ikey}=1)}(s, t, t_1, v) dt_1 \quad (1)$$

This requirement uses only the distributed Fortran code and is the primary numerical test in this note (Table 4).

Notation follows the code: s, t, v are the Mandelstam variables and the inelasticity, $u = v - s - t + 4m^2$, $\lambda_s = s(s - 4m^2)$ (`als`), m^2 is the electron mass squared (`m2`), $z_2 = z - s - t_1 + 4m^2$, and $D = s + t_1 - 4m^2 = 1/dz$.

1.2 Summary of faults

Integral	Line Block	Line with Issue	MERADGEN as-is	Corrected
aj47	309–316	310	(v-v)	(v-u)
aj29	233–239	234	+8d0*m2)	+8d0*m2**2)
		238	/2d0/m2	/2d0/m2**2
aj28	230–232	231	/2d0/m2	/2d0/m2**2
aj51	334–342	339	))+ (2d0*(	))- (2d0*(

Table 1: Faults in the **ikey=2** branch of `fsir.f`. Line numbers refer to the original Fortran source. The sign of the logarithmic term of `aj28` (line 230) is treated separately in Sec. 3.

Integral (variable in Fortran)	Definition (ikey=0)	Line Number	Used to Calculate
aj28	tt1*aj5	73	sr1 (358), sr5 (559)
aj29	tt1**2*aj5	74	sr1 (357)
aj47	m2*dz**3*aj8	92	sr10 (817)
aj51	dz**2*aj9	96	sr10 (811)

Table 2: Definitions and uses of the affected integrals (original Fortran line numbers).

2 Code faults

2.1 aj47: transcription error (line 310)

```
aj47=m2*(2d0*aj45*(s**2*(u-v)**2+6d0*m2**2*v**2-2d0*m2*s*(t*u
+2d0*v*(v-v)))-pi*(v*(-(s**3*u**2*v*(s**2-3d0*u**2+10d0*u*v
```

The factor (v-v) is identically zero and should read (v-u).

Justification. `aj47` is the integral of $m^2/(D^3 z_2)$ over the photon phase space. After the z -integration (which reproduces the code's own `ikey=1` form `aj8=-pi/sqrt(cz2)`) and the substitution $y = 1/D$, the t_1 -integral reduces exactly to a multiple of `aj45` plus algebraic boundary terms. The multiple is fixed by the coefficients of `cz2` (from `zd`, `meradgen10.f` lines 815–834). Rewriting those coefficients with the kinematic identity $s + t + u = v + 4m^2$, i.e. $s + t - 4m^2 = v - u$, yields the coefficient of `aj45` in the form printed in the code, with the term $2v(v - u)$ in place of $2v(v - v)$.

Numerical confirmation. With the correction, the closed form agrees with the numerical integral of the code's own `ikey=1` expression over the code's own t_1 limits to better than 10^{-40} at four kinematic points. MERADGEN as-is deviates by 0.03–1.7% at the test points of Table 4.

Structural cross-check. Under exchange of the two initial electrons at fixed detected electron, which is the $t \leftrightarrow u$ relation between the terms S_1 and S_{10} of Ref. [2] (in equation for S_{10}), the `sr1` term $-4(2m^2 - s)^2 \text{aj29}$ maps to the `sr10` group $-4(2m^2 - s)^2 [\text{aj51} + \text{aj31} - 2 \text{aj47} + 2 \text{aj27}]$. The group arises from partial fractions of $m^2/(z^2 z_2^2)$; its four coefficients match the code exactly. The implied identity, group at (s, t, v) equal to `aj29` at (s, u, v) , holds to 10^{-39} with the corrections of this note and fails at the 10^{-4} – 10^{-7} level with `(v-v)`.

Magnitude. At beam energies of 1 and 11 GeV, $\theta_{\text{CM}} = 90^\circ$, and $v/(s + t)$ between 10^{-3} and 10^{-1} , the correction changes `aj47` by 6×10^{-8} to 7×10^{-5} relative.

2.2 `aj29`: dimensional errors (lines 234, 238)

```
.+v**2*(2d0*s*t*v*(s+2d0*m2)-v**2*((s-2d0*m2)**2+8d0*m2)
...
.dlog((s-2d0*m2)*(s-2d0*m2+dsqrt(als))/2d0/m2
.-1d0))
```

Line 234. $(s - 2m^2)^2$ has mass dimension four; `8d0*m2` has dimension two. The correct term is `8d0*m2**2`.

Line 238. The argument of a logarithm must be dimensionless. The product $(s - 2m^2)(s - 2m^2 + \sqrt{\lambda_s})$ has dimension four, so it must be divided by m^4 , not m^2 . With m^4 , the argument is the square of a dimensionless ratio and the logarithm is

$$\ell_s = 2 \ln \frac{s - 2m^2 + \sqrt{\lambda_s}}{2m^2} = 2 \ln \frac{1 + \xi_s}{1 - \xi_s}, \quad \xi_s = \frac{\sqrt{s - 4m^2}}{\sqrt{s}}. \quad (2)$$

This is the logarithm used, without approximation, in Ref. [3, Eqs. (47), (54), (60)]. Its ultrarelativistic limit is $\ln(s^2/m^4) = s L_s$ with L_s of Ref. [2, Eq. (A11)]. The same logarithm is coded correctly in `aj21` (line 204, divided by `m2**2`) and, in simplified form, in `aj36` (line 268). In MERADGEN as-is, the argument depends on the unit system: in GeV the logarithm is shifted by $\ln m^2 \approx -15.2$.

Numerical confirmation. As-is, `aj29` deviates from the integrated `ikey=1` expression by up to 1.6%; only the combination of both corrections restores agreement (Table 4).

2.3 aj28: dimensional error (line 231)

```

    aj28=m2*pi/t**2/als*((s*(v-t)-2d0*m2*v)*
    .dlog((s-2d0*m2)*(s-2d0*m2+dsqrt(als))/2d0/m2
    .-1d0)/dsqrt(als)-(t*(s*v-als)-2d0*m2*v**2)/m2/v)

```

Line 231. The logarithm is the same ℓ_s as in aj29, with the same missing power of m^2 . aj28 and aj29 are both built on aj5 (lines 73–74), and the identical slip suggests one expression was copied from the other.

Remaining discrepancy. Correcting the logarithm does not by itself reconcile aj28 with its definition; see Sec. 3.

Scope. aj28 also enters sr5 (line 559), a t – u interference term.

2.4 aj51: sign error (line 339)

```

    --s*u*v))+(2d0*(-3d0*(m2**2+(s-m2)**2)*u*v**2+3d0*m2*(s-2d0*m2)

```

The sign preceding the logarithmic term should be negative.

Analytic justification. aj51 is the integral of $m^2/(D^2 z_2^2)$; aj45 is the integral of $1/(D z_2)$ over the same region. The derivation uses only definitions contained in the as-is Fortran code.

1. The z -integral of $1/z_2$ is $-\pi/\sqrt{c_{z2}}$ with $c_{z2} = a_z D^2 + 2b_z D + c_z$. Differentiating with respect to D gives the z -integral of $1/z_2^2$ as $\pi b_{z2}/c_{z2}^{3/2}$, using the exact identity $b_{z2} = a_z D + b_z$. This reproduces the code's `ikkey=1` form `aj9=m2*pi*bz2/cz2**1.5`.
2. Writing $c_{z2} = \alpha D^2 + \beta D + \gamma$, $b_{z2} = pD + q$, and substituting $y = 1/D$ gives

$$\text{aj45} = \pi \int \frac{dy}{\sqrt{Q}}, \quad \text{aj51} = -\pi m^2 \int \frac{y^2(p + qy)}{Q^{3/2}} dy, \quad Q = \gamma y^2 + \beta y + \alpha, \quad (3)$$

with identical limits and orientation.

3. The second integral reduces exactly to an algebraic term plus $c \int dy/\sqrt{Q}$, where the constant c follows from a linear system and equals $2K/(\lambda_s^2 u^3)$, with K the polynomial that multiplies the logarithm in the code. Hence

$$\text{aj51} = -\frac{2m^2 K}{\lambda_s^2 u^3} \text{aj45} + \text{algebraic terms}. \quad (4)$$

4. The code's logarithmic term, rewritten with its own aj45 (line 305), equals $+2m^2 K \text{aj45}/(\lambda_s^2 u^3)$: correct magnitude, **opposite sign**.

The most direct test is Eq. (1): the `ikkey=2` closed form must equal the `ikkey=1` expression integrated over t_1 between the code's limits. Table 3 gives this ratio for aj51.

Separating the exact integral into parts gives a ratio of exactly +1 for the algebraic part and exactly –1 for the logarithmic part. As $m^2 \rightarrow 0$ at fixed s, t, v , the absolute error falls roughly as m^2 , from 8×10^{-2} at $m^2 = 10^{-3}$ to 3×10^{-7} at $m^2 = 10^{-9}$.

			$R = \frac{\text{aj51}^{(\text{ik}ey=2)}(s, t, v)}{\int_{t_1^{\min}}^{t_1^{\max}} \text{aj51}^{(\text{ik}ey=1)} dt_1}$	
Chosen (s, t, v, m^2)	Rough E/m	$\sim \theta_{\text{CM}}$	MERADGEN as-is	Sign corrected
(20, -5, 1, 1)	9	68°	0.30	$1 - 2 \times 10^{-51}$
(10, -3, 0.5, 0.2)	24	70°	0.47	$1 - 3 \times 10^{-51}$
(8, -2, 0.7, 0.3)	12	66°	0.54	$1 + 4 \times 10^{-51}$
(1, -0.4, 0.05, 10^{-3})	499	79°	0.88	$1 + 2 \times 10^{-51}$

Table 3: Ratio of the `aj51` closed form to the `ik=1` expression integrated over t_1 between the code’s limits (`meradgen10.f` lines 113–115). Test points are in arbitrary units, with m^2 chosen comparable to s to expose the mass-dependent terms; they do not correspond to physical kinematics. $E/m = (s - 2m^2)/(2m^2)$ is the equivalent beam energy in the target rest frame, and $\cos \theta_{\text{CM}} = 1 + 2t/(s - 4m^2)$. For comparison, a physical electron beam has $E/m \approx 2 \times 10^3$ at 1 GeV and 2.2×10^4 at 11 GeV.

2.5 Why Faults Were Not Detected

- Every faulty term carries an explicit factor of m^2 and vanishes in the ultrarelativistic limit. In that limit `sr1` and `sr10` reproduce S_1 and S_{10} of Ref. [2] to 10^{-30} relative (checked at $m^2 = 10^{-30}$, $P = 0, \pm 1$), so comparison with the published formulas cannot see these errors.
- The two dimensional errors are invisible whenever $m^2 = 1$ in the chosen units.
- The built-in tests of Ref. [1, Sec. 6] compare generated distributions with cross sections computed by the same routine and are insensitive to errors in the routine itself.

3 Open item: sign of the logarithmic term in `aj28` (line 230)

```
aj28=m2*pi/t**2/als*((s*(v-t)-2d0*m2*v)*
```

With the line-231 correction applied, `aj28` still disagrees with the integral of the code’s own `ik=1` expression over the code’s own t_1 limits. In MERADGEN as-is, the deviation is -5.0% , -1.9% , -6.9% , and -0.12% at the four points of Table 4. The discrepancy is present at $m^2 = 1$, where the line-231 correction has no effect, so it is independent of the dimensional error.

Observations to date.

- Exact integration of the defining integrand, by the same reduction used for `aj51` (Sec. 2.4) with $y = 1/t_1$, reproduces the algebraic term of `aj28` exactly.
- The logarithmic term of the exact result has the same magnitude as the code’s and the opposite sign, at every point tested.
- Reversing the sign of the logarithmic term, together with the line-231 correction, gives agreement better than 3×10^{-51} at five kinematic points.

Open questions.

- The sign determination for `aj28` rests on a single analytic route and the self-consistency comparison; the independent two-dimensional integration over the Gram-determinant region has not been applied to `aj28`.
- A written derivation of the `aj28` logarithmic coefficient, analogous to Sec. 2.4, has not been completed; the substitution $y = 1/t_1$ involves $y < 0$, and the orientation conventions require explicit treatment.
- `aj28` enters both `sr1` (line 358) and `sr5` (line 559); the effect of the discrepancy on either term has not been quantified.

The candidate correction is

$$\text{aj28} = m_2^2 \pi / t^{**2} / \text{als} * (-(s*(v-t) - 2d0*m_2*v) *$$

pending resolution of these points.

4 Verification methods and cited inputs

Inputs from the code. The analytic arguments use only: the `ikkey=0` definitions (`fsir.f` lines 43–96); the z -quadratic coefficients from `zd` (`meradgen10.f` line 815; a_z at 821, b_{z2} at 832, c_{z2} at 834), together with the exact identities $c_{z2} = a_z D^2 + 2b_z D + c_z$ and $b_{z2} = a_z D + b_z$; the t_1 limits (`meradgen10.f` lines 113–115); and λ_s (`als`, `meradgen10.f` line 272).

Methods.

1. Self-consistency: numerical integration of the code’s own `ikkey=1` expressions over the code’s own t_1 limits, compared with the `ikkey=2` closed forms.
2. Independent integration: direct two-dimensional integration of the `ikkey=0` integrands over the physical region defined by the Gram determinant $\Delta(k_1, k_2, p_1, k)$.
3. Exact antiderivatives by reduction to $\int dy/\sqrt{Q}$, validated against methods 1 and 2 to 10^{-49} .
4. Ultrarelativistic comparison of `sr1` and `sr10` with S_1 and S_{10} of Ref. [2], using the logarithms of its Eq. (A11).

Numerical work used 30 to 80 significant digits; algebraic identities were checked symbolically. Test points use m^2 comparable to s deliberately, to expose mass-suppressed terms.

Coverage. Fifteen of the 51 integrals have been explicitly checked: `aj1`; `aj5`; `aj8`; `aj9`; `aj27`; `aj28`; `aj29`; `aj31`; and `aj45` thru `aj51`. The remaining 36 have not been explicitly reviewed; this would be an excellent project for a future student.

5 Constants

5.1 Which Were in Need of Updating

Electron mass squared (`meradgen10.f` line 268). The Fortran code’s implementation of m_e^2 (`m2/.261112d-6/`) is inconsistent with (`m/.511000d-3/`), whose square is 0.261121×10^{-6} ; the

Integral	MERADGEN as-is				Corrected
	P_1	P_2	P_3	P_4	
aj1, aj5, aj8, aj9	ϵ	ϵ	ϵ	ϵ	–
aj27, aj31	ϵ	ϵ	ϵ	ϵ	–
aj45, aj46, aj48-aj50	ϵ	ϵ	ϵ	ϵ	–
aj28	-5.0×10^{-2}	-1.9×10^{-2}	-6.9×10^{-2}	-1.2×10^{-3}	open (Sec. 3)
aj29	ϵ	-4.8×10^{-3}	-1.6×10^{-2}	-8.7×10^{-4}	ϵ
aj47	1.2×10^{-2}	4.5×10^{-3}	1.7×10^{-2}	2.6×10^{-4}	ϵ
aj51	–0.70	–0.53	–0.46	–0.12	ϵ

Table 4: Relative deviation of the `ikey=2` closed form from the `ikey=1` expression integrated over t_1 between the code’s limits. Kinematic points (s, t, v, m^2) : $P_1 = (20, -5, 1, 1)$; $P_2 = (10, -3, 0.5, 0.2)$; $P_3 = (8, -2, 0.7, 0.3)$; and $P_4 = (1, -0.4, 0.05, 10^{-3})$. ϵ denotes agreement better than 10^{-37} , the precision of the comparison. The corrected column applies Table 1.

last two digits are transposed. The existence of this issue was noted by Parker Zhiss prior to his graduation and cessation of his project. Since s is built from both `m` and `m2`, the kinematics are slightly inconsistent. The port computes m^2 from m and uses the currently-accepted PDG value.

Lepton masses in `vacpol` (meradgen10.f line 366). Updated per PDG. In MERADGEN as-is, the electron entry (`.26110d-6`) differs from line 268, so the code carries two different electron masses, and the τ entry corresponds to $m_\tau \approx 1.784$ GeV, a pre-1992 value.

A listing of all constant changes can be found in Table 5.

Constant	MERADGEN (Fortran)	C++ port	Notes
π	<code>atan(1d0)*4d0</code>	<code>std::acos(-1.0)</code>	Identical to double precision.
α	0.729735×10^{-2}	0.00729735256	$\alpha^{-1} = 137.03605$ (Fortran); 137.035999 (port).
m_e (GeV)	0.511000×10^{-3}	$0.51099895 \times 10^{-3}$	PDG value in port.
m_e^2 (GeV ²)	0.261112×10^{-6}	$m_e \cdot m_e = 2.6111993 \times 10^{-7}$	Fortran value $\neq m_e \cdot m_e$ (0.261121×10^{-6}); transposed digits.
m_e^2 in <code>vacpol</code> (GeV ²)	0.26110×10^{-6}	$m_e \cdot m_e$	Fortran carries a second electron mass ($m_e = 0.510979$ MeV).
m_μ^2 (GeV ²)	0.111637×10^{-1}	0.0111636912	$m_\mu = 105.6584$ MeV (Fortran); 105.65837 MeV (port).
m_τ^2 (GeV ²)	3.18301	3.1574802	$m_\tau = 1.78410$ GeV (Fortran, pre-1992); 1.77693 GeV (port).
$(\hbar c)^2$ (GeV ² nb)	0.389379×10^6	0.3893793721×10^6	Cross sections in nb. Current PDG: 0.3893793721×10^6 .
Hadronic VP A, B, C	<code>vacpol</code> , lines 374–384	as-is	Retained.

Table 5: Physical constants in MERADGEN 1.0 and in the C++ port. Fortran line numbers refer to `meradgen10.f`.

5.2 Needs to be Reviewed

Hadronic vacuum polarization (meradgen10.f lines 374–384). Retained as-is; however, someone (perhaps me later on) should review and correct. At polarimeter kinematics, $Q^2 \approx 5.6 \times 10^{-3} \text{ GeV}^2$ at 11 GeV and $\theta_{\text{CM}} = 90^\circ$, and the contribution is of order 10^{-4} in the cross section. However, it’s not clear to me this carries heavily into the analyzing power and so I’ve left this as needs to be reviewed; my guess is that this isn’t critical but we can should probably adjust these values just to say we’re as confident as possible in the accuracy of our calculations.

Range (GeV ²)	A (aaa)	B (bbb)	C (ccc)	Lines
$ t < 1$	-1.345×10^{-9}	-2.302×10^{-3}	4.091	374–376
$1 \leq t < 64$	-1.512×10^{-3}	-2.822×10^{-3}	1.218	378–380
$ t \geq 64$	-1.1344×10^{-3}	-3.0680×10^{-3}	0.99992	382–384

Table 6: Hadronic vacuum-polarization parameters in `vacpol`, $\text{Re } \Pi_h \simeq A + B \ln(1 + C|t|)$. Line numbers refer to the original Fortran source, `meradgen10.f`.

6 Conclusion

The corrections that I mentioned in this note were made for correctness and precision, not because the MERADGEN calculations are in any way greatly compromised. The errors noted are confined to a few calculated coefficients, the `ajNN` terms, while the underlying physics and the code’s defining expressions appear to be sound. All of these terms carry an explicit factor of the electron mass squared and vanish when the electron mass can be neglected. MERADGEN’s exact expressions have, however, been used at beam energies as low as 1.1 GeV and Q^2 down to $2 \times 10^{-4} \text{ GeV}^2$ [3], where $m^2/|t| \sim 10^{-3}$; the size of these corrections at such kinematics has not been established here. As energy decreases and one moves away from the ultrarelativistic regime, the impact of these terms on the analytic integrations becomes less clear. The corrections ensure that the C++ port matches its defining integrals at all kinematics.

7 Additional Notes

Ref. [2], Figs. 4–7, shows results at 1 GeV, labeled “JLab”. In particular, Fig. 5 shows the Born and radiatively corrected Møller asymmetries. These plots may be useful for comparison in the future. These plots may be relevant for comparison in the future.

References

- [1] A. Afanasev, E. Chudakov, A. Ilyichev, and V. Zykunov, “MERADGEN 1.0: Monte Carlo generator for the simulation of radiative events in parity conserving doubly-polarized Møller scattering,” *Comput. Phys. Commun.* **176**, 218 (2007), arXiv:hep-ph/0603027.
- [2] A. Ilyichev and V. Zykunov, “Lowest order QED radiative corrections to longitudinally polarized Møller scattering,” *Phys. Rev. D* **72**, 033018 (2005).

- [3] I. Akushevich, H. Gao, A. Ilyichev, and M. Meziane, “Radiative corrections beyond the ultra relativistic limit in unpolarized ep elastic and Møller scatterings for the PRad experiment at Jefferson Laboratory,” *Eur. Phys. J. A* **51**, 1 (2015).